

1. $(x^a)' =$
2. $(a^x)' =$
3. $(e^x)' =$
4. $(\log_a x)' =$
5. $(\ln |x|)' =$
6. $(\sin x)' =$
7. $(\cos x)' =$
8. $(\tan x)' =$
9. $(\cot x)' =$
10. $(\sec x)' =$
11. $(\csc x)' =$
12. $(\ln |\cos x|)' =$
13. $(\ln |\sin x|)' =$
14. $(\ln |\sec x + \tan x|)' =$
15. $(\ln |\csc x - \cot x|)' =$
16. $(\ln |\csc x + \cot x|)' =$
17. $(\arcsin x)' =$
18. $(\arccos x)' =$
19. $(\arctan x)' =$
20. $(\operatorname{arccot} x)' =$
21. $[\ln(x + \sqrt{x^2 + 1})]' =$
22. $[\ln(x + \sqrt{x^2 - 1})]' =$
23. $(\frac{1}{2} \arcsin x + \frac{x}{2} \sqrt{1 - x^2})' =$
24. $x^k \, dx =$
25. $\frac{1}{x} \, dx =$
26. $e^x \, dx =$
27. $a^x \, dx =$
28. $\sin x \, dx =$
29. $\cos x \, dx =$
30. $\tan x \, dx =$
31. $\cot x \, dx =$
32. $\frac{1}{\cos x} \, dx =$
33. $\frac{1}{\sin x} \, dx =$
34. $\frac{1}{\cos^2 x} \, dx =$
35. $\frac{1}{\sin^2 x} \, dx =$
36. $\sec x \tan x \, dx =$
37. $\csc x \cot x \, dx =$
38. $\frac{1}{1 + x^2} \, dx =$
39. $\frac{1}{a^2 + x^2} \, dx =$
40. $\frac{1}{\sqrt{1 - x^2}} \, dx =$
41. $\frac{1}{\sqrt{a^2 - x^2}} \, dx =$
42. $\sqrt{1 - x^2} \, dx =$
43. $\sqrt{a^2 - x^2} \, dx =$
44. $\frac{1}{\sqrt{x^2 + 1}} \, dx =$
45. $\frac{1}{\sqrt{x^2 + a^2}} \, dx =$
46. $\frac{1}{\sqrt{x^2 - 1}} \, dx =$
47. $\frac{1}{\sqrt{x^2 - a^2}} \, dx =$
48. $\sqrt{1 + x^2} \, dx =$
49. $\frac{1}{x^2 - a^2} \, dx =$
50. $\frac{1}{a^2 - x^2} \, dx =$
51. $\sin^2 x \, dx =$
52. $\cos^2 x \, dx =$
53. $\tan^2 x \, dx =$
54. $\cot^2 x \, dx =$
55. $\tan^n x \, dx =$

$$56. \int_0^{\frac{\pi}{2}} \sin^n x \, dx =$$

$$57. \int_0^{\frac{\pi}{2}} \cos^n x \, dx =$$

$$58. \int_0^{\pi} \sin^n x \, dx =$$

$$59. \int_0^{\pi} \cos^n x \, dx =$$

$$1. \sin\left(\frac{\pi}{2} \pm \alpha\right) =$$

$$2. \sin(\pi \pm \alpha) =$$

$$3. \cos\left(\frac{\pi}{2} \pm \alpha\right) =$$

$$4. \cos(\pi \pm \alpha) =$$

$$5. \sin(\alpha \pm \beta) =$$

$$1. e^x =$$

$$2. \sin x =$$

$$3. \cos x =$$

$$4. \arctan x =$$

$$5. \frac{1}{1-x} =$$

$$6. \frac{1}{1+x} =$$

$$7. \ln(1+x) =$$

$$8. (1+x)^a =$$

$$9. \tan x =$$

$$10. \arcsin x =$$

$$59. \int_0^{2\pi} \sin^n x \, dx =$$

$$60. \int_0^{2\pi} \cos^n x \, dx =$$

$$61. \int e^{ax} \cos bx \, dx =$$

$$62. \int e^{ax} \sin bx \, dx =$$

$$6. \cos(\alpha \pm \beta) =$$

$$7. \tan(\alpha \pm \beta) =$$

$$8. \tan\left(\frac{\pi}{4} - \alpha\right) =$$

$$9. \cot(\alpha \pm \beta) =$$

$$10. \sin 2\alpha =$$

$$11. \cos 2\alpha =$$

$$12. \sin 3\alpha =$$

$$13. \cos 3\alpha =$$

$$14. \sin^2 \frac{\alpha}{2} =$$

$$15. \cos^2 \frac{\alpha}{2} =$$